

Naji Mushtaha

DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

Contact

Phone: +970567148559
Email: naji.m.mushtaha@gmail.com
Address: Gaza city

LinkedIn: www.linkedin.com/in/naji-mushtaha

Professional Summary

Results-driven Data Science & AI Engineering student with a strong foundation in Python programming and data analysis. Proficient in developing algorithms and handling complex datasets, complemented by advanced skills in UI/UX and Graphic Design to effectively visualize data insights. Passionate about leveraging machine learning and analytical tools to build efficient, data-backed solutions. Led the AI engineering of VibeLocate AI, a team-built Dubai real-estate platform combining LLM-based natural-language search with geospatial sentiment scoring.

Technical Projects

[AI Engineer & Team Lead] VibeLocate AI | Dubai real-estate platform, 8-person team | 2026

- Built a Python/FastAPI service that turns Arabic and English search queries into structured filters using LLMs, with Pydantic validation and defensive output checks.
- Designed the “Vibe Score”: Haversine geospatial analysis (500 m radius) over 6,200+ OpenStreetMap points of interest, combined with review sentiment analysis, to rate safety, quietness and amenities from 0 to 10.
- Collected and cleaned the data: 6,200+ points of interest via the Overpass API (435 unnamed records removed), 700+ property records parsed from SQL dumps, and 720 AI-generated reviews clearly labeled as synthetic.
- Added reliability and monitoring: model fallback, retry with backoff, graceful degradation, a per-call metrics endpoint, and unit tests (pytest).
- Built a city-wide Vibe Heatmap batch pipeline with a per-run sentiment cache that removes duplicate LLM calls.
- Deployed the service on Render, documented the APIs, and integrated it with the team’s Laravel backend and Vue frontend.

[Data Science & Python Developer]

- **Data Analysis:** I have executed projects for data analysis using Python and libraries (Pandas, NumPy, Matplotlib), where I worked on data cleaning and extracting insights that helped in making data-driven decisions.
- **Databases:** I designed and managed databases using SQL, including building tables, creating complex queries, and ensuring efficient data relationships.
- **Programming and Development:** I developed software algorithms to solve logical and engineering problems, focusing on writing clean and scalable code.

[Graphic & UI/UX Designer]

- **Graphic Design:** I completed a range of advanced designs (visual identities, logos, social media posts) using Adobe Photoshop and Illustrator, with emphasis on academic design principles.
- **User Experience (UI/UX):** I designed application and website interfaces focused on enhancing user experience, linking visual aesthetics with technical usability.

Skills

Programming & Data: Python (Pandas, NumPy, Matplotlib), SQL
AI / ML: LLM integration & prompt engineering, NLP, sentiment analysis, machine learning foundations
Backend & Tools: FastAPI, Pydantic, REST APIs, pytest, Git/GitHub, Postman
Geospatial: OpenStreetMap / Overpass API, Haversine distance analysis
Design: Adobe Photoshop, Illustrator, UI/UX

Academic History

University College of Applied Sciences | 2023 – 2028
Master of Marketing
Relevant Coursework: Algorithms & Data Structures · Database Management Systems (SQL) · Statistical Data Analysis · Machine Learning Foundations
Cumulative GPA: 80.1

Awards and Certification

Python Programming Certificate | 2026
Focusing on building algorithms, data structures, and using software libraries in data analysis.

Advanced Graphic Design | 2026
Mastering advanced visual design using Adobe programs, and building visual identities for tech projects.

ICDL | 2025
Mastering basic digital skills and data management using the Microsoft Office suite.